Photon Momentum and the Compton Effect

PHYS 2203 — Introductory Physics III

October 21, 2026

Overview

The last two lectures, on blackbody radiation and the photoelectric effect, showed that light carries energy in discrete photons of energy $E = hf$. A particle also carries momentum. This lecture argues that a photon of wavelength λ carries momentum $p = h/\lambda$, and tests this with the Compton effect, in which X-rays scattered from electrons come out with a longer wavelength that depends on the scattering angle. Treating the scattering as a collision between a photon and an electron, with the relativistic energy and momentum of Lectures 17–19, gives the shift exactly.

1 Photon Momentum

The photoelectric effect shows that a photon carries energy $E = hf$. A particle also carries momentum. Two independent arguments suggest what a photon's momentum should be.

1. **Special relativity.** A photon travels at c, so it is massless, and Lecture 19 showed that a massless particle has $E = pc$.

2. **Classical electromagnetism.** Electromagnetic waves carry momentum, which is the origin of radiation pressure. For a wave, the momentum per unit volume equals the energy per unit volume divided by c, again $p = E/c$.

Combining $p = E/c$ with $E = hf = hc/\lambda$,

$$\boxed{p = \frac{E}{c} = \frac{hf}{c} = \frac{h}{\lambda}.} \tag{1}$$

The first direct test of this was the Compton effect.

2 The Compton Effect

The experiment

In 1923 Arthur Compton directed **X-rays**, electromagnetic radiation with wavelengths of roughly 0.01 to 10 nm, much shorter than visible light, at a carbon target. The X-rays had a single wavelength λ, and he measured the wavelength of the X-rays scattered at various angles θ (Figure 1). The outer electrons of carbon are so weakly bound, compared with the X-ray energies, that they behave as free electrons at rest.

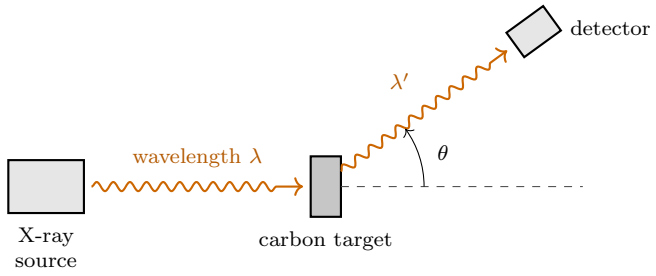


Figure 1: The Compton experiment. X-rays of wavelength λ strike a carbon target, and a detector measures the wavelength λ' of X-rays scattered at angle θ.

Classical prediction. An electromagnetic wave makes a free electron oscillate at the wave's frequency, and the oscillating electron radiates at that same frequency. Classically, the scattered radiation should therefore have the same wavelength as the incident radiation, at every angle.

Observation. At each angle the scattered X-rays contain two wavelengths (Figure 2): one equal to the incident wavelength λ, and one at a *longer* wavelength λ'. The shift $\lambda' - \lambda$ grows with the scattering angle, and it does not depend on the target material or on the incident wavelength.

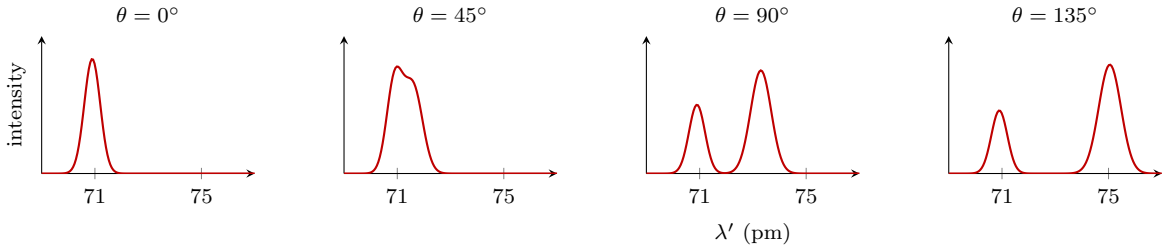


Figure 2: Wavelengths of X-rays scattered from carbon at four angles, for incident wavelength $\lambda = 70.9$ pm (schematic, after Compton's data). Besides the peak at the incident wavelength, a second peak appears at a longer wavelength that moves farther out as θ increases.

The photon explanation: a two-particle collision

Compton treated the scattering as a collision between a single photon and a single electron at rest (Figure 3). The photon arrives with energy $E = hc/\lambda$ and momentum $p = h/\lambda$. Afterward the photon leaves at angle θ with energy $E' = hc/\lambda'$ and momentum $p' = h/\lambda'$, and the electron recoils at angle ϕ with energy E_e and momentum p_e. The recoiling electron can move fast, so we use the relativistic expressions of Lectures 17 and 18 for it.

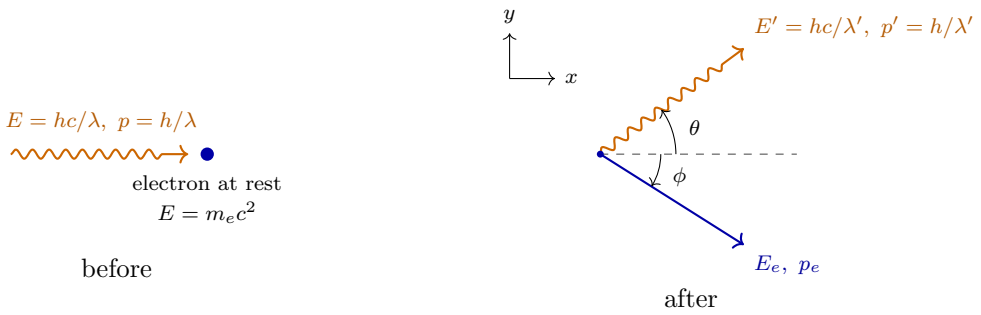


Figure 3: Compton scattering as a collision. A photon strikes an electron at rest. Afterward the photon scatters at angle θ and the electron recoils at angle ϕ.

Conservation of energy.

$$E + m_e c^2 = E' + E_e \qquad \Longrightarrow \qquad E_e = E - E' + m_e c^2. \tag{A}$$

Conservation of momentum. In components,

$$x\text{:}\quad p = p'\cos\theta + p_e \cos\phi, \tag{2}$$
$$y\text{:}\quad 0 = p'\sin\theta - p_e \sin\phi. \tag{3}$$

These are three equations in four unknowns (p', θ, p_e, ϕ). The photon can scatter at any angle θ, but once θ is chosen, the other three are fixed. We want λ' in terms of θ, so we eliminate the electron's angle ϕ and then its momentum.

Step 1: eliminate ϕ. Move the photon terms to the other side and square each component:

$$p_e^2 \cos^2\phi = (p - p'\cos\theta)^2, \tag{4}$$
$$p_e^2 \sin^2\phi = (p'\sin\theta)^2. \tag{5}$$

Adding the two equations and using $\cos^2\phi + \sin^2\phi = 1$ and $\cos^2\theta + \sin^2\theta = 1$,

$$p_e^2 = p^2 - 2pp'\cos\theta + p'^2. \tag{B}$$

Step 2: use the energy–momentum relation for the electron. From Lecture 18, $E_e^2 = p_e^2 c^2 + m_e^2 c^4$. Substitute (A) on the left and (B) on the right, and use $pc = E$ and $p'c = E'$ for the photons:

$$\left(E - E' + m_e c^2\right)^2 = E^2 - 2EE'\cos\theta + E'^2 + m_e^2 c^4. \tag{6}$$

Expanding the left side,

$$E^2 + E'^2 + m_e^2 c^4 - 2EE' + 2(E - E')m_e c^2 = E^2 - 2EE'\cos\theta + E'^2 + m_e^2 c^4. \tag{7}$$

The terms E^2, E'^2, and $m_e^2 c^4$ cancel, leaving

$$(E - E')\, m_e c^2 = EE'(1 - \cos\theta). \tag{8}$$

Step 3: convert to wavelengths. Divide both sides by $EE'm_ec^2$:

$$\frac{1}{E'} - \frac{1}{E} = \frac{1}{m_ec^2}(1 - \cos\theta). \tag{9}$$

With $1/E = \lambda/hc$ and $1/E' = \lambda'/hc$, multiplying through by hc gives the **Compton formula**:

$$\boxed{\lambda' - \lambda = \frac{h}{m_ec}(1 - \cos\theta).} \tag{10}$$

Comparison with experiment

The constant

$$\frac{h}{m_ec} = \frac{hc}{m_ec^2} = \frac{1240 \text{ eV} \cdot \text{nm}}{511\,000 \text{ eV}} = 0.00243 \text{ nm} = 2.43 \text{ pm} \tag{11}$$

is called the **Compton wavelength** of the electron. The formula agrees with Compton's measurements.

* The shift depends only on θ, not on λ or on the target, as observed.
* Since $1 - \cos\theta \geq 0$, the scattered photon always has a longer wavelength, and so less energy. The energy it loses becomes the kinetic energy of the recoiling electron, $K_e = hc/\lambda - hc/\lambda'$.
* The shift is largest for backscattering, $\theta = 180°$, where $\lambda' - \lambda = 2h/m_ec = 4.85$ pm.
* The unshifted peak comes from photons scattering off a tightly bound inner electron, so that the whole atom recoils. The shift is then h/Mc, with M the mass of the atom, which is too small to see.

Compton received the 1927 Nobel Prize. His experiment showed that photons carry momentum h/λ as well as energy hf, and that in a collision they behave exactly like particles.